

MATÍAS FERNÁNDEZ LAKATOS

Data Scientist · Machine Learning Engineer · Research Engineer

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mflakatos.github.io/MFernandezLakatos.github.io/

PROFESSIONAL SUMMARY

Physicist with postgraduate training (Ph.D. and MSc in Physics, MSc in Big Data Analytics) currently working as a Research Engineer at Gradient, developing intelligent systems for early anomaly and threat detection in corporate cybersecurity. Expertise in unsupervised machine learning, real-time large-scale data processing, and explainable AI (XAI). Over 9 years of experience in research and teaching, with a strong background in statistical modeling, computer vision, and data-driven decision making. Neo4j Certified Professional with hands-on experience in graph databases and graph data science.

PROFESSIONAL EXPERIENCE

Research Engineer

Gradient · Vigo, Spain · 2025 – present

- Development of intelligent solutions for early threat detection in corporate cybersecurity using unsupervised ML models.
- Design and implementation of β -Variational Autoencoder, Autoencoder, Extended Isolation Forest and Isolation Forest models for real-time anomaly detection on large-scale data.
- Focus on model explainability (XAI) to enhance usability for non-technical users and improve trust in AI systems.
- Integration into scalable data architecture using Kafka, Spark and Airflow for efficient real-time stream processing.
- Experiment tracking and model versioning with MLFlow; large artifact storage with MinIO.

Researcher, Lecturer & Teaching Assistant

Faculty of Engineering, Universidad de la República (Udelar) · Montevideo, Uruguay · 2015 – 2024

- Taught university-level physics; developed and delivered course materials for undergraduate students.
- Research in QCD long-range properties, multispectral instruments for remote monitoring, phase object retrieval, and novel integration methods.
- Published 4 first-author peer-reviewed articles in Scopus-indexed journals and 2 co-authored articles.
- Presented research at international conferences (SPIE, IOP, Elsevier).

EDUCATION

MSc in Big Data Analytics Technologies

Universidad de Santiago de Compostela (USC) · Santiago de Compostela, Spain · 2024 – 2025

Thesis: Application of new anomaly detection models in Big Data contexts. Courses: Large-Scale Databases, Data Mining, Data Visualization, Business Intelligence, IoT, Statistics.

Ph.D. in Optics (Physics)

Universidad de la República (Udelar) · Montevideo, Uruguay · 2019 – 2024

Thesis: Visualization and Characterization of Phase Objects. Key courses: Reinforcement Learning, Computer Image Processing, Computational Multivariate Statistics, Deep Learning for CV (CS231n). Advisors: PhD. Ferrari, PhD. Frins, PhD. Ayubi.

MSc in Quantum Chromodynamics (Theoretical Physics)

Universidad de la República (Udelar) · Montevideo, Uruguay · 2016 – 2018

Thesis: Role of the various couplings in infrared QCD. Courses: Statistical Mechanics, QFT I & II, General Relativity. Advisors: PhD. Wschebor, PhD. Peláez.

Bachelor's Degree in Physics

Universidad de la República (Udelar) · Montevideo, Uruguay · 2011 – 2015

CERTIFICATIONS

	Certification	Issuer	Date
■	Neo4j Certified Professional	Neo4j GraphAcademy	May 2026
■	Master Python for DevOps and CI/CD	Udemy	2025
■	Using Neo4j with Python	Neo4j GraphAcademy	May 2026
■	Intermediate Cypher Queries	Neo4j GraphAcademy	May 2026
■■■	Importing Data Fundamentals	Neo4j GraphAcademy	May 2026
■	Graph Data Modeling Fundamentals	Neo4j GraphAcademy	May 2026
■	Cypher Fundamentals	Neo4j GraphAcademy	Apr 2026
■■■	Neo4j Fundamentals	Neo4j GraphAcademy	Apr 2026

Full Neo4j profile: graphacademy.neo4j.com/u/34d98b08-78b3-4f35-a97c-3c869a7dec4e/

TECHNICAL SKILLS

Machine Learning & AI: scikit-learn, PyTorch, TensorFlow, Keras, MLFlow | Unsupervised Learning, Anomaly Detection, Autoencoders, Variational Autoencoders, Isolation Forest | XAI, Reinforcement Learning, Deep Learning, Computer Vision

Data Engineering & Big Data: Apache Kafka, Apache Spark, Apache Airflow, MinIO | Real-time Stream Processing, Large-Scale Pipelines, ETL, IoT Data

Graph Databases: Neo4j, Cypher, Graph Data Modeling, Graph Data Science | Knowledge Graphs, Network Analysis

Programming & Tools: Python, R, SQL, MATLAB, LaTeX | Git, Linux, Docker (basic)

Data Analysis & Visualization: pandas, NumPy, SciPy, matplotlib, seaborn, plotly | Statistical Analysis, Multivariate Analysis, Optimization

Computer Vision & Image Processing: OpenCV, scikit-image, PIL/Pillow | Phase Object Visualization, Digital Image Processing, Pattern Recognition

PUBLICATIONS

First-Author (Peer-Reviewed, Scopus Indexed)

1. [2024 – SPIE] Integration with one partial derivative applied to quantitative phase imaging
2. [2023 – OLE, Scopus] Common-path quantitative phase imaging by propagation through a sinusoidal intensity mask
3. [2022 – Optik, Scopus] Phase retrieval by amplifying the prism term of the Transport of Intensity Equation
4. [2019 – IJMPA, Scopus] On the contribution of different coupling constants in the infrared regime of Yang-Mills theory

Co-Authored

1. [2025 – OLT] Wavefront reconstruction of discontinuous phase objects using directional derivatives
2. [2024 – IOP Science] Hilbert's and Takeda's single-shot interferometry with a linear-carrier: a comparison

ACHIEVEMENTS & FELLOWSHIPS

- Member of the National System of Researchers (SNI), Uruguay, 2024
- PhD Scholarship, Comisión Académica de Posgrados (CAP), 2022
- PhD Scholarship, Agencia Nacional de Investigación e Innovación (ANII), 2019
- Master's Scholarship, Comisión Académica de Posgrados (CAP), 2016

LANGUAGES

Spanish — Native | English — Professional working proficiency